

Tetrahedral sums of Hermitian matrices

Thomas C. Fraser¹, Matthias Christandl¹, and Anton Alekseev²

¹Department of Mathematical Sciences, University of Copenhagen, Universitetsparken 5, 2100 Denmark

²[TODO: add Anton's affiliations here]

August 2, 2024

Abstract

TODO: write an abstract

1 Introduction

This is the main document for this project. Previous drafts are located in the folder titled “old” and will not necessarily compile properly.

2 Tetrahedra & Hermitian matrices

2.1 Definition

2.2 Symmetries

2.3 Special cases

3 Non-linear inequalities

Lemma 1. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices, then*

$$\sum_j \lambda_j(A+B)^2 + \sum_j \lambda_j(B+C)^2 \leq \sum_j \lambda_j(B)^2 + \sum_j \lambda_j(A+B+C)^2. \quad (1)$$

Proof. Note that $\sum_j \lambda_j(X)^2 = \text{Tr}[X^2]$ for any Hermitian operator X . Therefore,

$$\begin{aligned} & \text{Tr}[(B)^2] + \text{Tr}[(A+B+C)^2] - \text{Tr}[(A+B)^2] - \text{Tr}[(B+C)^2] \\ &= (\text{Tr}[B^2]) + (\text{Tr}[A^2] + \text{Tr}[B^2] + \text{Tr}[C^2] + 2\text{Tr}[AB] + 2\text{Tr}[BC] + 2\text{Tr}[AC]) \\ & \quad - (\text{Tr}[A^2] + \text{Tr}[B^2] + 2\text{Tr}[AB]) - (\text{Tr}[B^2] + \text{Tr}[C^2] + 2\text{Tr}[BC]) \\ &= 2\text{Tr}[AC]. \end{aligned} \quad (2)$$

Since $A, C \geq 0$ are both positive semidefinite, we also have $\text{Tr}[AC] \geq 0$ which proves the claim. $\square$

Theorem 2. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices, then*

$$\sum_j \lambda_j(A+B) \log \lambda_j(A+B) + \sum_j \lambda_j(B+C) \log \lambda_j(B+C) \quad (3)$$

$$\leq \sum_j \lambda_j(B) \log \lambda_j(B) + \sum_j \lambda_j(A+B+C) \log \lambda_j(A+B+C). \quad (4)$$

- 3.1 Proof via spectral embedding
- 3.2 Proof via representation theory
- 4 Conclusion

References

- [AB68] Vishnu K Agrawala and Johan G Belinfante. “Graphical formulation of recoupling theory for any compact group”. In: *Annals of Physics* 49.1 (1968), pp. 130–170.
- [AH11] Daiki Akimoto and Masahito Hayashi. “Discrimination of the change point in a quantum setting”. In: *Physical Review A* 83.5 (2011), p. 052328.
- [AHP10] Koenraad Audenaert, Fumio Hiai, and Dénes Petz. “Strongly subadditive functions”. In: *Acta Mathematica Hungarica* 128.4 (2010), pp. 386–394.
- [Ais97] AK Aiston. “A skein theoretic proof of the hook formula for quantum dimension”. In: *arXiv preprint q-alg/9711019* (1997).
- [Alc+23] Judith Alcock-Zeilinger et al. “Wigner 6j symbols for SU (N): Symbols with at least two quark-lines”. In: *Journal of Mathematical Physics* 64.2 (2023).
- [Alc18] Judith M Alcock-Zeilinger. *The Special Unitary Group, Birdtracks, and Applications in QCD*. 2018.
- [Ale+18] Anton Alekseev et al. “Concentration of symplectic volumes on Poisson homogeneous spaces”. In: *arXiv preprint arXiv:1808.06975* (2018).
- [Ale97] A Yu Alekseev. “On Poisson actions of compact Lie groups on symplectic manifolds”. In: *Journal of Differential Geometry* 45.2 (1997), pp. 241–256.
- [AM07] Anton Alekseev and Eckhard Meinrenken. “Ginzburg-Weinstein via Gelfand-Zeitlin”. In: *Journal of Differential Geometry* 76.1 (2007), pp. 1–34.
- [AM98] AK Aiston and HR Morton. “Idempotents of Hecke algebras of type A”. In: *Journal of Knot Theory and Its Ramifications* 7.04 (1998), pp. 463–487.
- [AMS20] Victor Alekseev, Andrey Morozov, and Alexey Sleptsov. “Multiplicity-free $U_q(\mathfrak{sl}_N)$ 6-j symbols: relations, asymptotics, symmetries”. In: *Nuclear Physics B* 960 (2020), p. 115164.
- [AMW00] Anton Alekseev, Eckhard Meinrenken, and Christopher Woodward. “Group-valued equivariant localization”. In: *Inventiones mathematicae* 140.2 (2000), pp. 327–350.
- [AMW01] Anton Alekseev, Eckhard Meinrenken, and Chris Woodward. “Linearization of Poisson actions and singular values of matrix products”. In: *Annales de l’institut Fourier*. Vol. 51. 6. 2001, pp. 1691–1717.
- [AMW99] Anton Alekseev, Eckhard Meinrenken, and Chris Woodward. “Duistermaat-Heckman distributions for group valued moment maps”. In: *arXiv preprint math/9903087* (1999).
- [APS17a] Anton Alekseev, Masha Podkopaeva, and Andras Szenes. “A symplectic proof of the Horn inequalities”. In: *Advances in Mathematics* 318 (2017), pp. 711–736.
- [APS17b] Anton Alekseev, Masha Podkopaeva, and Andras Szenes. “The Horn problem and planar networks”. In: *Advances in Mathematics* 318 (2017), pp. 678–710.
- [Aqu+12] Vincenzo Aquilanti et al. “Semiclassical mechanics of the Wigner 6j-symbol”. In: *Journal of Physics A: Mathematical and Theoretical* 45.6 (2012), p. 065209.
- [Art23] DV Artamonov. “Classical-symbols of finite-dimensional representations of the algebra”. In: *Theoretical and Mathematical Physics* 216.1 (2023), pp. 909–923.
- [Art24] Dmitry Artamonov. “Calculation of 6j-symbols for the Lie algebra $\mathfrak{gl}_n$ ”. In: *arXiv preprint arXiv:2405.05628* (2024).
- [AW97] Sharad Agnihotri and Chris Woodward. “Eigenvalues of products of unitary matrices and quantum Schubert calculus”. In: *arXiv preprint alg-geom/9712013* (1997).
- [Bac10] Miriam Backens. “The 6j symbols and an extended version of Horn’s problem”. In: (2010).
- [Bar03] John W Barrett. “Geometrical measurements in three-dimensional quantum gravity”. In: *International Journal of Modern Physics A* 18.suppl02 (2003), pp. 97–113.
- [BDS11] Eugenio Bianchi, Pietro Dona, and Simone Speziale. “Polyhedra in loop quantum gravity”. In: *Physical Review D* 83.4 (2011), p. 044035. arXiv: [1009.3402v2](#).

- [Bel02] Prakash Belkale. “Geometric proofs of Horn and saturation conjectures”. In: *arXiv preprint math/0208107* (2002).
- [Bel04] Prakash Belkale. “Invariant theory of $GL(n)$ and intersection theory of Grassmannians”. In: *International Mathematics Research Notices* 2004.69 (2004), pp. 3709–3721.
- [Bel08] Prakash Belkale. “Quantum generalization of the Horn conjecture”. In: *Journal of the American Mathematical Society* 21.2 (2008), pp. 365–408.
- [BGS08] Mustapha Bouhtou, Stéphane Gaubert, and Guillaume Sagnol. “Optimization of network traffic measurement: a semidefinite programming approach”. In: *Proceedings of the International*. Vol. 230. 2008.
- [Bha01] Rajendra Bhatia. “[Linear algebra to quantum cohomology: the story of Alfred Horn’s inequalities](#)”. In: *The American Mathematical Monthly* 108.4 (2001), pp. 289–318.
- [BK16] Prakash Belkale and Shrawan Kumar. “The multiplicative eigenvalue problem and deformed quantum cohomology”. In: *Advances in Mathematics* 288 (2016), pp. 1309–1359.
- [BN23] Bruce Bartlett and Nzaganya Nzaganya. “Coherent loop states and angular momentum”. In: *arXiv preprint arXiv:2306.17293* (2023).
- [Boa07] Philip P Boalch. “[Regge and Okamoto symmetries](#)”. In: *Communications in mathematical physics* 276.1 (2007), pp. 117–130.
- [BVW18] Nicole Berline, Michèle Vergne, and Michael Walter. “The Horn inequalities from a geometric point of view”. In: *L’Enseignement Mathématique* 63.3 (2018), pp. 403–470.
- [Cha10] Laurent Charles. “On the quantization of polygon spaces”. In: (2010).
- [Chr08] Matthias Christandl. “A Quantum Information-Theoretic Proof of the Relation between Horn’s Problem and the Littlewood-Richardson Coefficients”. In: *Lecture Notes in Computer Science*. Springer Berlin Heidelberg, 2008, pp. 120–128.
- [CMZ19] Robert Coquereaux, Colin McSwiggen, and Jean-Bernard Zuber. “Revisiting Horn’s problem”. In: *Journal of Statistical Mechanics: Theory and Experiment* 2019.9 (2019), p. 094018.
- [CMZ20] Robert Coquereaux, Colin McSwiggen, and Jean-Bernard Zuber. “On Horn’s problem and its volume function”. In: *Communications in Mathematical Physics* 376.3 (2020), pp. 2409–2439.
- [Coq23] Robert Coquereaux. “About integer-valued variants of the theta and $6j$ symbols”. In: *Journal of Mathematical Physics* 64.3 (2023).
- [CP95] Vyjayanthi Chari and Andrew N Pressley. *A guide to quantum groups*. Cambridge university press, 1995.
- [CŞW18] Matthias Christandl, M Burak Şahinoğlu, and Michael Walter. “Recoupling coefficients and quantum entropies”. In: *Annales Henri Poincaré*. Vol. 19. 2. Springer. 2018, pp. 385–410.
- [CW06] Nicola Ciccoli and Pawel Witkowski. “From Poisson to quantum geometry”. In: *Lecture Notes*, available at <http://toknotes.mimuw.edu.pl> (2006).
- [CZ18] Robert Coquereaux and Jean-Bernard Zuber. “From orbital measures to Littlewood–Richardson coefficients and hive polytopes”. In: *Annales de l’Institut Henri Poincaré D* 5.3 (2018), pp. 339–386.
- [DH05] Sumit Daftuar and Patrick Hayden. “[Quantum state transformations and the Schubert calculus](#)”. In: *Annals of Physics* 315.1 (2005), pp. 80–122.
- [Esp12] Chiara Esposito. “On the classical and quantum momentum map”. In: (2012). 1203.4112: [arxiv](#).
- [Feg78] HD Fegan. “The heat equation on a compact Lie group”. In: *Transactions of the American Mathematical Society* 246 (1978), pp. 339–357.
- [FG13] S Friedland and S Gaubert. “Submodular spectral functions of principal submatrices of a hermitian matrix, extensions and applications”. In: *Linear Algebra and its Applications* 438.10 (2013), pp. 3872–3884.

- [FK00] Laurent Freidel and Kirill Krasnov. “Simple spin networks as Feynman graphs”. In: *Journal of Mathematical Physics* 41.4 (2000), pp. 1681–1690.
- [FKL10] Laurent Freidel, Kirill Krasnov, and Etera R Livine. “Holomorphic factorization for a quantum tetrahedron”. In: *Communications in Mathematical Physics* 297.1 (2010), pp. 45–93.
- [FL03] Laurent Freidel and David Louapre. “Asymptotics of 6j and 10j symbols”. In: *Classical and Quantum Gravity* 20.7 (2003), p. 1267.
- [FNR07] L Freidel, K Noui, and Ph Roche. “6J symbols duality relations”. In: *Journal of mathematical physics* 48.11 (2007).
- [FT14] Peter J Forrester and Colin J Thompson. “The Golden-Thompson inequality: Historical aspects and random matrix applications”. In: *Journal of Mathematical Physics* 55.2 (2014).
- [Ful00] William Fulton. “Eigenvalues, invariant factors, highest weights, and Schubert calculus”. In: *Bulletin of the American Mathematical Society* 37.3 (2000), pp. 209–249.
- [Ful97] William Fulton. *Young tableaux: with applications to representation theory and geometry*. 35. Cambridge University Press, 1997.
- [Gur08] Razvan Gurau. “The Ponzano–Regge Asymptotic of the 6 j Symbol: An Elementary Proof”. In: *Annales Henri Poincaré*. Vol. 9. Springer. 2008, pp. 1413–1424.
- [Hes20] Sarah Tanja Hess. “Examination of the connection between the Horn Problem and the Lax Conjecture”. In: (2020).
- [HHR16] Hal M Haggard, Muxin Han, and Aldo Riello. “Encoding curved tetrahedra in face holonomies: phase space of shapes from group-valued moment maps”. In: *Annales Henri Poincaré*. Vol. 17. Springer. 2016, pp. 2001–2048.
- [HK05] Andre Henriques and Joel Kamnitzer. *Crystals and coboundary categories*. 2005. arXiv: [math/0406478 \[math.QA\]](#).
- [HK06] André Henriques and Joel Kamnitzer. “The octahedron recurrence and gl_n crystals”. In: *Advances in Mathematics* 206.1 (2006), pp. 211–249.
- [Hor62] Alfred Horn. “Eigenvalues of sums of Hermitian matrices.” In: (1962).
- [Hub20] Felix Huber. *Positive Maps and Matrix Contractions from the Symmetric Group*. 2020. arXiv: [2002.12887](#).
- [HW22] Felix Huber and Nikolai Wyderka. *Refuting spectral compatibility of quantum marginals*. 2022. arXiv: [2211.06349v2](#).
- [Kly00] Alexander A Klyachko. “Random walks on symmetric spaces and inequalities for matrix spectra”. In: *Linear Algebra and its Applications* 319.1-3 (2000), pp. 37–59.
- [Knu00] Allen Knutson. “The symplectic and algebraic geometry of Horn’s problem”. In: *Linear Algebra and its Applications* 319.1-3 (2000), pp. 61–81.
- [KPS24] Stefan Keppeler, Simon Plätzer, and Malin Sjodahl. “Wigner 6j symbols with gluon lines: completing the set of 6j symbols required for color decomposition”. In: *Journal of High Energy Physics* 2024.5 (2024), pp. 1–25.
- [KR90] AN Kirillov and N Yu Reshetikhin. “Representations of the Algebra $U_q(\mathfrak{sl}(2))$, q-Orthogonal Polynomials and Invariants of Links”. In: *New developments in the theory of knots* 11 (1990), p. 202.
- [KTW03] Allen Knutson, Terence Tao, and Christopher T Woodward. “A positive proof of the Littlewood–Richardson rule using the octahedron recurrence”. In: *arXiv preprint math/0306274* (2003).
- [Lam05] McKenzie Lamb. “A proof of the Thompson conjecture using Poisson geometry”. In: (2005).
- [LE03] Jiang-Hua Lu and Sam Evens. “Thompson’s conjecture for real semi-simple Lie groups”. In: (2003). arXiv: [math/0310098](#).
- [LS23] Elena Lanina and Alexey Sleptsov. “Tug-the-hook symmetry for quantum 6j-symbols”. In: *Physics Letters B* 845 (2023), p. 138138.

- [McS21] Colin McSwiggen. “The Harish-Chandra integral: an introduction with examples”. In: *L’Enseignement Mathématique* 67.3 (2021), pp. 229–299.
- [MMS17] A Mironov, A Morozov, and A Sleptsov. “On 6 j-symbols for symmetric representations of $U_q(\mathfrak{su}(N))$ ”. In: *JETP Letters* 106 (2017), pp. 630–636.
- [Moa84] Daniel S Moak. “The q-analogue of Stirling’s formula”. In: *The Rocky Mountain Journal of Mathematics* (1984), pp. 403–413.
- [Mou84] J Moussouris. “Quantum models of space-time based on recoupling theory”. PhD thesis. University of Oxford, 1984.
- [MR05] Annalisa Marzuoli and Mario Rasetti. “Computing spin networks”. In: *Annals of Physics* 318.2 (2005), pp. 345–407.
- [MS18] Andrey Morozov and Alexey Sleptsov. “New Symmetries for the $U_q(\mathfrak{sl}(N))$ 6-j Symbols from the Eigenvalue Conjecture¹”. In: *JETP Letters* 108 (2018), pp. 697–704.
- [PR68] Giorgio Ponzano and Tullio Eugenio Regge. “Semiclassical limit of Racah coefficients”. In: *Spectroscopic and group theoretical methods in physics* (1968).
- [Res87] N Yu Reshetikhin. *Quantized universal enveloping algebras, the Yang-Baxter equation and invariants of links, 1*. Tech. rep. 1987.
- [Ric02] G Richardson. “The trigonometry of the tetrahedron”. In: *The Mathematical Gazette* 2.32 (1902), pp. 149–158.
- [Rob02] Justin Roberts. “Asymptotics and 6j-symbols”. In: *Geom. Topol. Monogr* 4 (2002), pp. 245–261.
- [Rob99] Justin Roberts. “Classical 6j-symbols and the tetrahedron”. In: *Geometry & Topology* 3.1 (1999), pp. 21–66.
- [RR86] David P Robbins and Howard Rumsey Jr. “Determinants and alternating sign matrices”. In: *Advances in Mathematics* 62.2 (1986), pp. 169–184.
- [RW00] Tullio Regge and Ruth M Williams. “Discrete structures in gravity”. In: *Journal of Mathematical Physics* 41.6 (2000), pp. 3964–3984.
- [SBT17] David Sutter, Mario Berta, and Marco Tomamichel. “Multivariate trace inequalities”. In: *Communications in Mathematical Physics* 352.1 (2017), pp. 37–58. arXiv: [1604.03023v3](#).
- [Spe05] David E Speyer. “Horn’s problem, Vinnikov curves, and the hive cone”. In: (2005).
- [TT73] RC Thompson and S Therianos. “The Eigenvalues and Singular Values of Matrix Sums and Products. VII (1)”. In: *Canadian Mathematical Bulletin* 16.4 (1973), pp. 561–569.
- [TW06] Yuka U Taylor and Christopher T Woodward. “6 j symbols for and non-Euclidean tetrahedra”. In: *Selecta Mathematica* 11.3-4 (2006), p. 539.
- [VMK88] Dmitrii Aleksandrovich Varshalovich, Anatolij Nikolaevič Moskalev, and Valerii Kel’manovich Khersonskii. *Quantum theory of angular momentum*. World Scientific, 1988.
- [Zub18] Jean-Bernard Zuber. “Horn’s problem and Harish-Chandra’s integrals. Probability density functions”. In: *Annales de l’Institut Henri Poincaré D* 5.3 (2018), pp. 309–338.

5 [Research Notes]

5.1 Horn’s problem & generalizations

- Horn’s original paper where they conjectured the form of a necessary and sufficient set of inequalities via an inductive scheme for increasing dimension [Hor62]
- An excellent expository overview of the big picture by Knutson [Knu00]
- An excellent introduction to these ideas is the recent masters thesis on various aspects of additive and multiplicative Horn problems in connection to Vinnikov curves and the Lax conjecture due to Hess and supervised by S. Kuhlmann and M. Infusino [Hes20]
- Quantum-info style proof of Littlewood-Richardson coefficient connection due to Christandl [Chr08]
- Coquereaux and Zuber explicitly outline the link between the orbital measures (i.e. the induced distributions in the sense of Klyachko’s random walk approach) and the magnitude of the stretched Littlewood-Richardson coefficients [CZ18]. See also the papers [CMZ20; CMZ19; Zub18].
- A recent article by McSwiggen is an excellent introduction to the Harish-Chandra integral [McS21]. The proof of the Harish-Chandra formula provide by McSwiggen using the heat equation methods from Fegan’s 1978 paper [Feg78] seems very promising for deriving a connection to the character formula for the norm of the recoupling operator!
- Speyer [Spe05]
- Generalizations of Horn’s problem to the product of unitary matrices [AW97; Bel08; BK16]
- Bhatia [Bha01]
- History account offered by Daftuar and Hayden [DH05], and technical review article by Fulton [Ful00].
- Berline, Vergne, and Walter a pedagogical account of the triple role played by Horn’s inequalities in characterizing the existence of invariant vectors in tensor products, the eigenvalues of sums of Hermitian matrices, and intersections of Schubert varieties [BVW18]
- Fulton provides a comprehensive treatment of the connection between representation theory of $GL(d)$ and the geometry and (co)homology of the complex Grassmannians [Ful97]
- Prakash Belkale has contributed greatly to the theory surrounding Horn-type problems: for example, in [Bel04], Belkale provides a *geometric* proof of the equivalence between (i) the structure coefficients of the representation theory of $GL(d)$ (i.e. the LR-coefficients $c_{\mu\nu}^\lambda$), and (ii) the structure coefficients of the cohomology ring of the Grassmannian $Gr(d, n)$ of d -dimensional subspaces in $\mathbb{C}^d$. This proof is now complementary to the previously known algebraic proofs which rely on knowledge of the algebra of symmetric functions. This geometric approach taken by Belkale, which emphasizes the tangent spaces of the involved Schubert varieties, is later used by Belkale to provide a geometric proof of the saturation conjecture [Bel02].

5.2 Thompson’s conjecture

- Original (indirect) proof through the lense of random walks Klyachko [Kly00]
- The proof of Thompson’s conjecture in the style of Alekseev, Meinrenken, and Woodward is the “linearization paper” due to [AMW01].
- See also the related article due to Lamb which summarizes the ideas of [AMW01] while also providing more technical background [Lam05]
- Lu and Evens extend the proof technique of [AMW01] to a larger class of Lie groups [LE03]

- Alekseev, Podkopaeva, and Szenes [APS17a]
- Alekseev, Podkopaeva, and Szenes [APS17b]
- Alekseev and Meinrenken [AM07]
- Deforming the symplectic volume by the s parameter Alekseev et al. [Ale+18]
- **TODO:** track down pdfs for the original triple of papers by Thompson and Therianos titled “On the singular values of matrix products” I, II, and III from 1973
- **TODO:** track down pdfs for the original triple of papers by Thompson and Freede titled “On the eigenvalues of sums of Hermitian matrices” I, II, and III from 1973
- **TODO:** track down a pdf for Thompson’s 1988 “Matrix Spectral Inequalities” referenced by Klyachko
- Thompson and Therianos [TT73] present a number of interesting counter-examples to the question of whether the inequalities pertaining to the spectra of general matrices can be deduced from looking at the spectra of diagonal matrices (both in the additive and multiplicative cases).

5.3 $6j$ symbols

- For a comprehensive reference textbook for the $6j$ coefficients, see the textbook by Varshalovich, Moskalev, and Khersonskii [VMK88], specifically Chapter 9.
- A great summary by Roberts [Rob02]; also the original result for the asymptotics of the $6j$ symbols due to Roberts [Rob99]
- The masters essay/thesis of Backens (**TODO: check if the pdf Matthias sent to me is the final version**) [Bac10]
- The spectral quantum marginal problem and the $6j$ symbol for the symmetric group Christandl, Şahinoğlu, and Walter [CSW18]
- Usage of Wigner’s $6j$ symbol to compute asymptotic error rates for the promise problem of distinguishing between two possible “change points” for a sequence of unknown pure states to transition from one unknown type to another [AH11]
- See also Coquereaux’s later paper which promotes the alternative, albeit equivalent, usage of the integer-valued gon and tet functions in place of the rational-valued theta (triangular) and irrational-valued $6j$ symbols [Coq23]. Most relevant for this project is [Coq23, Section IV D] which discusses the apparent lack of substantial generalization from $SU(2)$ to $SU(N)$ (“*apart from very few special cases (low dimension of irreps and no multiplicity) worked out by physicists at the time of the eightfold way, we are not aware of any published general result*”).
- The recent paper by Alekseev, Morozov, and Sleptsov (different Alekseev!) covers the case of some multiplicity free formulas for the quantum $6j$ symbols for $U_q(\mathfrak{sl}_N)$ [AMS20]
- Keppeler, Plätzer, and Sjodahl recently calculated $SU(d)$ $6j$ symbols with adjoint representations [KPS24] and a bit earlier [Alc+23] calculated $SU(d)$ $6j$ symbols with at least two fundamental representations [Alc+23].
- Representation theoretic proof of the Regge symmetries due to Boalch [Boa07]
- Asymptotics of quantum $6j$ symbols Taylor and Woodward [TW06]
- The integral formula for the square of the $SU(2)$ $6j$ symbol can be performed by an appropriate change of variables [FL03]. See also other papers by Laurent Freidel and collaborators that are relevant [FK00; FNR07; FKL10].

- Alcock-Zeilinger consider the case of $6j$ symbols for $SU(N)$ for wherein two opposing edges are labelled by fundamental irreps of $SU(N)$ so that all vertices are multiplicity one (or zero) [Alc18] TC: They also freely use the birdtrack notation which I enjoy.
- Only the tetrahedral symmetry of the $U_q(SL(N))$ $6j$ -symbol is known, the counterpart of the Regge symmetry is still unknown, although Morozov and Sleptsov conjectures the generalized Regge symmetry may be related to the eigenvalue conjecture, just as it is in the case when $N = 2$ [MS18].
- The graphical notion, specifically the use of an additional edge for multiplicity spaces can has been used for a long time, as early as [AB68; Mou84].
- The original Ponzano-Regge paper [PR68]
- Gurau derives the Ponzano-Regge asymptotic formula directly using Stirling’s approximation applied to Racah’s single-sum formula [Gur08]
- Bartlett and Nzaganya work with so-called *coherent loop states* or *Bohr-Sommerfeld states* to develop some semiclassical results concerning the inner products between such states (see also the talk on Youtube) [BN23]
- Aquilanti et al. provides an excellent introduction to subject of classical $6j$ symbols, their history, diagrammatics, presentations and asymptotics [Aqu+12]
- Marzuoli and Rasetti connects quantum computing models to the recoupling theory of $SU(2)$ angular momentum and also considers the semiclassical limit [MR05]
- Artamonov derives an explicit formula for the $d = 3$ case [Art23], and later to arbitrary d [Art24].
- There is a series of papers involving Alexey Sleptsov, starting with a computation of quantum $6j$ symbols for generic dimension d but with symmetric representations in terms of certain hypergeometric polynomials [MMS17], then with a discussion of the eigenvalue hypothesis and its potential connection to generalized Regge symmetries [MS18], then with a calculation of all multiplicity free quantum $6j$ symbols for arbitrary dimension [AMS20], and then finally with the discovery of novel “tug-the-hook” symmetries for quantum $6j$ symbols for all d [LS23].

5.4 Minkowski’s theorem and tetrahedron geometry

- Connecting group-valued moment maps to spherical, flat, and hyperbolic tetrahedron via Minkowski’s theorem from the reconstruction of flat tetrahedron Haggard, Han, and Riello. See also the comments in [HHR16, Section 11] on the utility of *quasi-poisson* (controlled failure of the Jacobi identity) manifolds and group-valued moment maps.
- Foundational results concerning the relationships between the edge lengths, face areas, dihedral and solid angles of a tetrahedron by Richardson [Ric02]
- Bianchi, Dona, and Speziale uses Minkowski’s theorem to parametrize convex polyhedra in $\mathbb{R}^3$ with specified face areas and the associated Kapovich-Millson phase space (i.e., the *space of shapes of polyhedra with fixed areas* and then considers its connection to loop quantum gravity via the usual Guillemin-Sternberg quantization commutes with reduction philosophy which associates invariants of the quantization of the unreduced phase space to the reduced phase space [BDS11] (notice the connection with [HHR16]).
- Regarding an algorithmic construction for the polyhedra appearing in Minkowski’s theorem and their complexity, see this post: [constructive-algorithm-for-minkowskis-theorem](#)
- Charles quantization of polygon spaces, tetrahedron and semiclassical limits [Cha10]

5.5 Quantization, Poisson geometry & quantum groups

- A popular textbook on quantum groups with an emphasis on the connection to Poisson manifolds, Lie bialgebras and their quantizations by Chari and Pressley [CP95]
- Course notes on Poisson geometry with an eye toward quantization by Ciccoli and Witkowski [CW06]
- The PhD thesis of Esposito (from the maths department here in Copenhagen) serves as a introduction to the world of Poisson momentum maps and the notion of a “quantum” momentum map as a deformation of the ordinary “classical” momentum map (where deformation is in the sense of deformation quantization) [Esp12].
- Alekseev [Ale97]
- Papers contained results pertaining to the generalization of Duistermaat-Heckman measures for Hamiltonian group actions with group-valued moment maps by Alekseev, Meinrenken, and Woodward [AMW99; AMW00]
- Kirillov and Reshetikhin [KR90]
- Octahedron Recurrence/Rule Robbins and Rumsey Jr [RR86] applied to the “Hive” ring Knutson, Tao, and Woodward [KTW03], and also Henriques and Kamnitzer [HK06]
- Henriques and Kamnitzer [HK05]
- A reference which contains the formula for the quantum dimension of a representation of $U_q(\mathfrak{sl}(d))$ labelled by λ , thanks Rinat! [Ais97]. See also [AM98] which clarifies the relation between various parameters. Apparently, the x, v, s parameters are apparently natural parameters to consider in the context of invariants of knots and links. The substitution mentioned in [AM98] is the following

$$s = e^{h/2}, \quad v = e^{-dh/2} = s^{-d}, \quad x = e^{-h/2d} = s^{-\frac{1}{d}}, \quad q = e^h = s^2.$$

This is consistent with [Res87] in the sense that

$$[k] := \frac{s^{+k} - s^{-k}}{s^{+1} - s^{-1}} = \frac{q^{+k/2} - q^{-k/2}}{q^{+1/2} - q^{-1/2}}. \quad (5)$$

- Reshetikhin develops the diagrammatics of the category of representations of $U_q(\mathfrak{g})$ motivated by connections to the developing theory of polynomial invariants of knots and links [Res87].
- Moak investigates a q -Stirling’s formula for the logarithm of the q -factorial [Moa84].

5.6 Eigenvalue inequalities

- The general subject of multivariate trace inequalities involving exponentials of sums of Hermitian and products of their exponentials is, of course, intimately related. See for instance the paper by Sutter, Berta, and Tomamichel [SBT17].
- Also the article [FT14] by Forrester and Thompson which describes the connection between Thompson’s inequality and hyperbolic geometry quite explicitly.
- Friedland and Gaubert considers submodular functions associated to operator monotones evaluated on different principal submatrices of a single Hermitian matrix [FG13]. While it is not clear to be yet whether or not one can relate partial sums of matrices to principal submatrices, [FG13] references a paper by Bouhtou, Gaubert, and Sagnol [BGS08] which proves the following family of inequalities for all positive semidefinite $A, B, C \geq 0$ (and $p \in [0, 1]$) [BGS08, Lemma 5.3]:

$$\sum_j \lambda_j (A + B)^p + \sum_j \lambda_j (B + C)^p \geq \sum_j \lambda_j (B)^p + \sum_j \lambda_j (A + B + C)^p. \quad (6)$$

- Audenaert, Hiai, and Petz consider a similar form of “strongly subadditive” functions [AHP10]
- For multivariate inequalities based on idempotents of the symmetric group algebra see Huber’s paper [Hub20].
- For spectral inequalities derived using semidefinite programming, see Huber and Wyderka’s paper [HW22].

5.7 Uncategorized

- Elementary introduction to toy models of three-dimensional quantum space [Bar03]
- Regge and Williams [RW00] a relatively recent review of discrete approaches to gravity. The essential idea, in contrast to the differential geometry of general relativity, is to model curvature discretely instead of continuously. In particular, one can think of a triangulation of the surface of a sphere; the curvature of the sphere is encoded in the triangulation when a configuration of three (or more) triangles meet at a common vertex with sum of interior angles deviating from π . It is in this sense that Regge and Williams refers to these types of models as “piecewise-flat”. Contained is an overview of various attempts to generalize discrete theories of three-dimensional gravity to four dimensions [RW00, Section III].